

TEAM -11
4/4 CSD – C
FINAL YEAR PROJECT SUBMISSION

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PROJECT GUIDE:
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Final-Year Project

1. MediAgent: An Agentic AI-Based Intelligent Patient Follow-Up and Healthcare Assistance System

1. Title:

MediAgent: An Agentic AI-Based Intelligent Patient Follow-Up and Healthcare Assistance System

Domain:

Primary Domain:

- Artificial Intelligence (AI)

Sub-Domains:

- Agentic AI, Healthcare Informatics, Generative AI (Large Language Models), Multi-Agent Systems, Digital Health, Medical Decision Support Systems, Remote Patient Monitoring

Abstract:

Abstract

MediAgent is an Agentic AI-based patient follow-up and healthcare assistance system designed to improve communication between doctors and patients after medical consultations. The platform enables patients who have already consulted a registered doctor to securely interact with the healthcare system remotely, reducing unnecessary hospital visits while ensuring continuous monitoring of their recovery. By leveraging Artificial Intelligence and Large Language Models (LLMs), the system provides personalized follow-up care, improves treatment adherence, and enhances the overall patient experience.

The proposed system utilizes multiple specialized AI agents, each responsible for a specific healthcare task. A **Patient Interaction Agent** collects daily health updates, a **Medication Agent** sends medication reminders and tracks adherence, a **Symptom Analysis Agent** evaluates changes in the patient's condition, a **Risk Detection Agent** identifies warning signs requiring immediate medical attention, and a **Summary Agent** generates concise health reports for doctors. These collaborative agents automate routine follow-up activities, assist healthcare professionals in monitoring patient progress, and support timely clinical decision-making.

The application features dedicated dashboards for both doctors and patients. Patients can access prescriptions, receive medication reminders, report symptoms, and communicate through an AI-assisted chat interface, while doctors can monitor patient recovery, review AI-generated summaries, track medication compliance, and identify high-risk patients through an intuitive dashboard. The system is developed using **React** for the frontend, **FastAPI (Python)** for the backend, **PostgreSQL/SQLite** for database management, and **LangGraph/CrewAI** for orchestrating multiple AI agents integrated with an LLM. By combining Agentic AI with remote patient monitoring, MediAgent offers an intelligent, efficient, and accessible healthcare solution that enhances patient engagement and supports informed clinical decisions.

2. Autonomous Data Analyst Agent: An AI-Powered Multi-Agent System for Automated Data Analysis and Insight Generation

2. Title:

Autonomous Data Analyst Agent: An AI-Powered Multi-Agent System for Automated Data Analysis and Insight Generation

Domain:

Artificial Intelligence (AI), Agentic AI, Data Science & Analytics, Machine Learning, Generative AI (LLMs)

Abstract:

Organizations generate massive amounts of data every day, but extracting meaningful insights often requires skilled data analysts and significant manual effort. This project proposes an **Autonomous Data Analyst Agent**, an intelligent AI system capable of performing end-to-end data analysis with minimal human intervention. The system leverages **Large Language Models (LLMs)** and **Agentic AI** principles to autonomously understand user queries, preprocess datasets, perform exploratory data analysis (EDA), generate visualizations, build predictive models when required, and provide actionable business insights.

The proposed solution employs multiple specialized AI agents, each responsible for a specific task such as data cleaning, statistical analysis, visualization, machine learning, and report generation. These agents collaborate to complete complex analytical workflows efficiently. Users simply upload a dataset in CSV or Excel format and interact with the system using natural language questions like *"What are the key trends?"*, *"Predict future sales"*, or *"Identify anomalies in the data."*

The system automatically detects data types, handles missing values, removes duplicates, selects appropriate analytical techniques, generates charts, summarizes findings, and produces comprehensive reports. By integrating technologies such as **LangGraph/CrewAI, LangChain, Python, Pandas, Scikit-learn, Matplotlib, FastAPI, and Streamlit**, the project demonstrates how autonomous AI agents can replace repetitive data analysis tasks while improving speed, accuracy, and accessibility.

This project showcases the practical application of Agentic AI in business intelligence and decision support systems, making advanced data analytics accessible even to users without technical expertise.

3. FinanceGenie: An Intelligent Personal Finance and Budget Planning Agent

3. Title:

FinanceGenie: An Intelligent Personal Finance and Budget Planning Agent

Domain:

Primary Domain:

- **Artificial Intelligence (AI)**

Sub-Domains:

- **Agentic AI, Personal Finance, FinTech, Generative AI (LLMs), Multi-Agent Systems, Data Analytics**

Abstract:

FinAgent is an Agentic AI-based personal finance management and financial advisory system designed to help individuals effectively manage their income, expenses, savings, and financial goals. The application enables users to securely upload bank statements or manually record financial transactions while interacting with an AI-powered assistant through natural language. By automating financial tracking and analysis, the system helps users gain better control over their spending habits, improve budgeting, and make informed financial decisions.

The system utilizes multiple specialized AI agents that collaborate to perform different financial tasks. An **Expense Tracking Agent** categorizes transactions automatically, a **Budget Planning Agent** creates personalized monthly budgets based on income and spending patterns, a **Savings Advisor Agent** recommends strategies to achieve financial goals, an **Investment Guidance Agent** provides educational investment suggestions based on user preferences and risk tolerance, an **Anomaly Detection Agent** identifies unusual spending behavior, and a **Financial Summary Agent** generates easy-to-understand reports and insights. Together, these agents provide intelligent, personalized financial assistance while reducing the complexity of personal money management.

The application includes an interactive dashboard where users can visualize expenses, monitor savings, set financial goals, and receive AI-generated recommendations through a conversational chat interface. The proposed system is developed using **React** for the frontend, **FastAPI (Python)** for the backend, **PostgreSQL or SQLite** for data storage, and **LangGraph/CrewAI** for orchestrating multiple AI agents integrated with a **Large Language Model (LLM)**. By combining Agentic AI with financial analytics, FinAgent delivers an intelligent and user-friendly personal finance solution that promotes responsible spending, encourages better financial planning, and empowers users to make smarter financial decisions.